52. Deep earth mining and drilling reveals the nature of rocks deep down the surface. **Volcanic eruption** forms another source of obtaining direct information. For example, volcanic eruptions and existence of hot springs, geysers etc. point to an interior which is very hot. Spatial variations in gravity, seismic waves, magnetic field, and meteors are indirect sources of information about Earth's interior.
53. The gravitation force (g) is not the same at different latitudes on the surface. It is greater near the poles and less at the equator. This is because of the distance from the centre at the equator being greater than that at the poles. Understanding Earth's interior helps us in understanding internal structure of various solar system objects as solar system is believed to be evolved from single source.
54. The high temperatures of Earth's core is attributed to automatic disintegration of the radioactive substances, i.e., decay of radioactive substances, whereas energy source for sun is nuclear fusion.
55. Abrupt release of energy along a fault causes earthquake waves. The point where the energy is released is called the focus of an earthquake, alternatively, it is called the hypocentre. The point on the surface, nearest to the focus, is called epicentre. It is the first one to experience the waves. It is a point directly above the focus.
56. All natural earthquakes take place in the lithosphere (depth up to 200 km from the surface of the earth). Earthquake waves are basically of two types — body waves and surface waves. Body waves are generated due to the release of energy at the focus and move in all directions travelling through the body of the earth. Hence, the name body waves.
57. Surface waves are long period waves and called as 'L' waves. They are low frequency, long wavelength and transverse vibration. Usually surface waves records higher on seismographs than that of primary waves or body waves and are responsible for most the destructive force of earthquake.